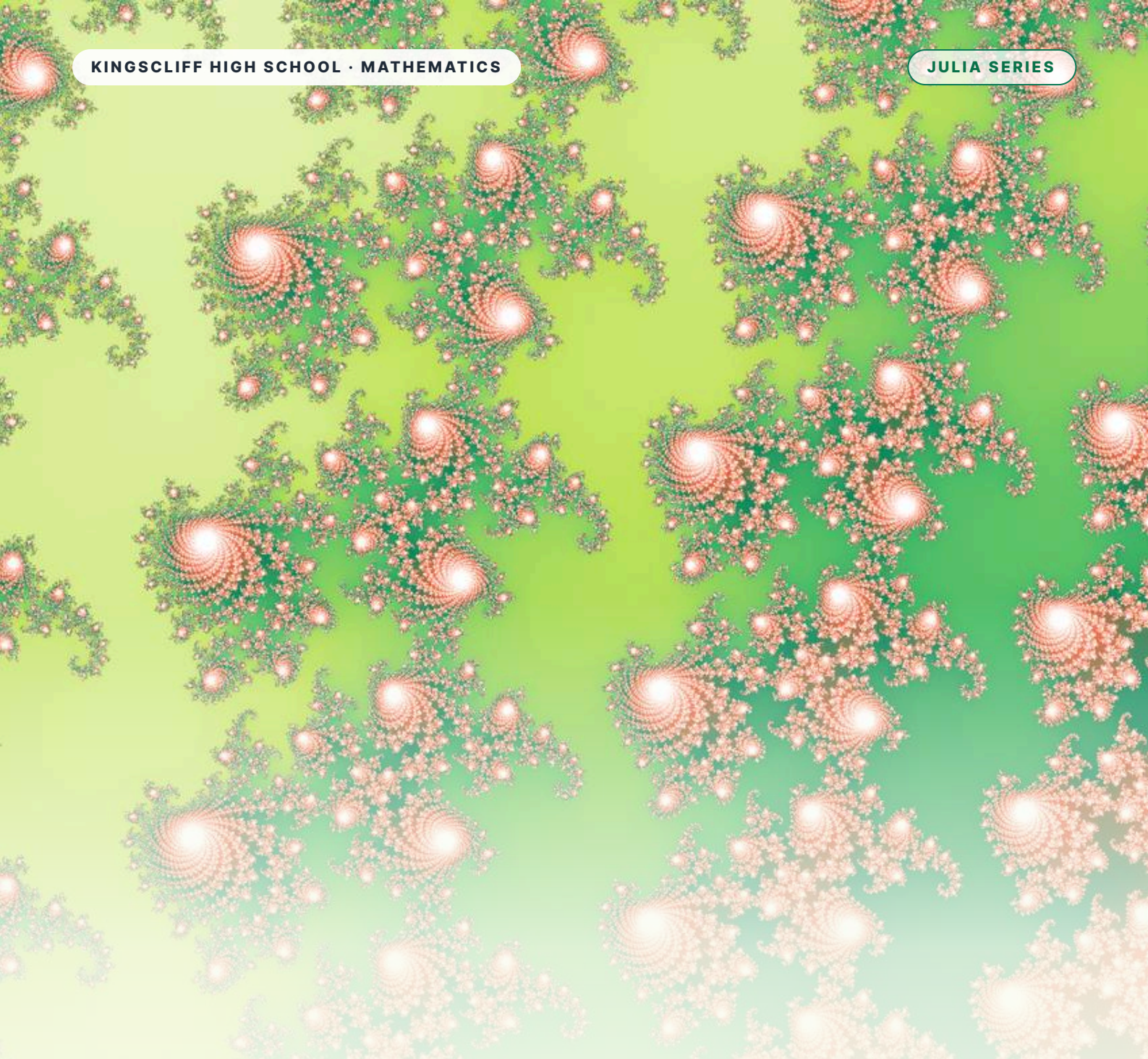


KINGSCLIFF HIGH SCHOOL · MATHEMATICS

JULIA SERIES

YEAR 10 ADVANCED → YEAR 11 MATHEMATICS ADVANCED

Summer Maths

Ready for Year 11

Six weeks. Twenty questions a week. Every Year 10 skill that Year 11 Mathematics Advanced builds on – mixed together so it sticks.

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6

NAME _____ CLASS _____

How it works

One page a week

Each week has **20 questions** and a **bonus**. Set aside about **40 minutes** – one sitting or a few short ones. Questions start easier and get a little harder through the page.

The same pattern every week

Question 1 is always indices, question 10 is always functions, and so on. If the same number keeps going wrong, that is the skill to revise before Year 11.

1	Indices	11	Graphs
2	Surds	12	Polynomials
3	Logarithms	13	Lines
4	Expand	14	Coordinate geometry
5	Factorise	15	Trig ratios & exact values
6	Algebraic fractions	16	Sine & cosine rules
7	Linear equations & inequalities	17	Geometry
8	Quadratic equations	18	Probability
9	Simultaneous equations	19	Financial maths
10	Functions	20	Problem solving

Why Julia? Each banner is part of the same *Julia set*: pick a number c, then repeat $z \to z^2 + c$ for every point. Points that stay small are coloured in. Every booklet in the series zooms into a different corner of the same set – this one is a deep zoom, about 35 times closer than the full set.

NSW syllabus (NESA Mathematics K–10, 2022). Revises MA5-IND-P-01, MA5-IND-P-02, MA5-LOG-P-01, MA5-ALG-P-01, MA5-ALG-P-02, MA5-EQU-P-01, MA5-EQU-P-02, MA5-FNC-P-01, MA5-NLI-P-01, MA5-POL-P-01, MA5-LIN-P-01, MA5-TRG-P-01, MA5-TRG-P-02, MA5-CIR-P-01, MA5-GEO-P-01, MA5-PRO-C-01, MA5-PRO-P-01, MA5-FIN-C-02. ★ Year 11 preview: Mathematics Advanced 11–12 (2024) focus areas Working with functions; Trigonometry and measure of angles; Trigonometric identities and equations; Introduction to differentiation.

Rules of the game

- **No calculator.** Leave answers exact (e.g. $2 \pm \sqrt{5}$ or $\frac{3\pi}{4}$) unless asked to round.
- **Show working** in the space under each question.
- **Mark it yourself** using the answers at the back (pages 10–12). Each answer has a line of working to help you find a mistake.
- **Fix it.** Redo anything you got wrong a day or two later, without looking.
- ★ questions preview Year 11 Mathematics Advanced (Stage 6). Have a go – they are the ones that will help most in Term 1.
- **Stuck?** The hints page (page 9) has the key facts and formulas.

My summer tracker

WEEK	DATE	SCORE	DONE	BONUS	REVISE (Q NUMBERS)
1 Week 1		/ 20	☐	☐	
2 Week 2		/ 20	☐	☐	
3 Week 3		/ 20	☐	☐	
4 Week 4		/ 20	☐	☐	
5 Week 5		/ 20	☐	☐	
6 Week 6		/ 20	☐	☐	

First up in Year 11 Mathematics Advanced: working with functions, graph transformations, trigonometry and measure of angles, exponential and logarithmic functions, and an introduction to differentiation. Every one of them starts from the skills in these six pages.

WEEK 1

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Indices, surds & logs · Algebra · Equations · Functions & graphs · Coordinate geometry · Trigonometry · Geometry & chance · In context · ★ Year 11 preview (Stage 6 Advanced) · No

1 INDICES

Evaluate $8^{\frac{2}{3}} + 4^{-\frac{1}{2}}$

2 SURDS

Simplify $\sqrt{75} - \sqrt{12}$

3 LOGARITHMS

Evaluate $\log_3 81$

4 EXPAND

Expand $(2x - 5)(x + 3)$

5 FACTORISE

Factorise $x^2 - 3x - 28$

6 ALGEBRAIC FRACTIONS

Simplify $\dfrac{x+2}{3} + \dfrac{x-1}{4}$

7 LINEAR EQUATIONS & INEQUALITIES

Solve $7 - 3x \geq 19$

8 QUADRATIC EQUATIONS

Solve $x^2 + 2x - 15 = 0$

9 SIMULTANEOUS EQUATIONS

Solve simultaneously: $y = 2x - 1$ and $x + y = 8$

10 FUNCTIONS

$f(x) = x^2 - 3x$. Find $f(-2)$.

11 GRAPHS

State the vertex and axis of symmetry of $y = (x - 2)^2 + 3$.

12 POLYNOMIALS

$P(x) = x^3 - 2x^2 + 4$. Find $P(2)$.

13 LINES

Find the gradient and y-intercept of $4x - 2y + 6 = 0$.

14 COORDINATE GEOMETRY

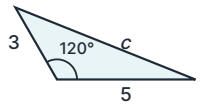
A is $(-2,\ 1)$ and B is $(4,\ 9)$. Find the midpoint and the length of AB.

15 TRIG RATIOS & EXACT VALUES

Find the exact value of $\sin 60° \times \tan 30°$

16 SINE & COSINE RULES

Find c.



17 GEOMETRY

$ABCD$ is a cyclic quadrilateral with $\angle A = 75°$. Find $\angle C$.

18 PROBABILITY

Two dice are rolled. Find $P(\text{the sum is at least } 10)$.

19 FINANCIAL MATHS

$5000 is invested at 4% p.a., compounded annually. Find its value after 2 years.

20 PROBLEM SOLVING

The product of two consecutive positive even numbers is 168. Find them.

BONUS If $x + \dfrac{1}{x} = 3$, find $x^2 + \dfrac{1}{x^2}$.

WEEK 2

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Indices, surds & logs · Algebra · Equations · Functions & graphs · Coordinate geometry · Trigonometry · Geometry & chance · In context · ★ Year 11 preview (Stage 6 Advanced) · No

1 INDICES

Simplify $(27x^6)^{\frac{1}{3}}$

2 SURDS

Rationalise the denominator of $\dfrac{10}{\sqrt{5}}$

3 LOGARITHMS

Evaluate $\log_{10} 0.001$

4 EXPAND

Expand and simplify $(x+1)(x^2-x+1)$

5 FACTORISE

Factorise $2x^2 - 8$

6 ALGEBRAIC FRACTIONS

Simplify $\dfrac{x^2-4x}{x^2-16}$

7 LINEAR EQUATIONS & INEQUALITIES

Solve $\dfrac{2x-1}{5} < 3$

8 QUADRATIC EQUATIONS

Solve $x^2 - 4x - 1 = 0$, giving exact answers.

9 SIMULTANEOUS EQUATIONS

Solve simultaneously: $2x + 3y = 13$ and $5x - 3y = 1$

10 FUNCTIONS

State the domain of $f(x) = \sqrt{x-3}$.

11 GRAPHS

Find the x- and y-intercepts of $y = x^2 - x - 6$.

12 POLYNOMIALS

Find the remainder when $P(x) = x^3 + 2x - 5$ is divided by $(x-1)$.

13 LINES

Find the equation of the line through $(3,\ -1)$ with gradient -2.

14 COORDINATE GEOMETRY

Find the centre and radius of $(x-1)^2 + (y+2)^2 = 16$.

15 TRIG RATIOS & EXACT VALUES

Solve $\sin\theta = \dfrac{\sqrt{3}}{2}$ for $0° \le \theta \le 180°$.

16 SINE & COSINE RULES

In $\triangle ABC$, $a = 6$, $A = 45°$ and $B = 30°$. Find b exactly.

17 GEOMETRY

Two similar solids have matching lengths in the ratio $2:3$. The smaller has volume 40 cm³. Find the volume of the larger.

18 PROBABILITY

A bag has 3 red and 2 white balls. Two are drawn without replacement. Find $P(\text{same colour})$.

19 FINANCIAL MATHS

A \$30 000 car depreciates by 15% p.a. Find its value after 2 years.

20 PROBLEM SOLVING

A rectangle has a perimeter of 26 cm and an area of 40 cm². Find its dimensions.

BONUS Show that $\dfrac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} = 4 + \sqrt{15}$.

WEEK 3

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Indices, surds & logs · Algebra · Equations · Functions & graphs · Coordinate geometry · Trigonometry · Geometry & chance · In context · ★ Year 11 preview (Stage 6 Advanced) · No

1 INDICES

Solve $9^x = 27$

2 SURDS

Expand and simplify $(2\sqrt{3} - 1)^2$

3 LOGARITHMS

Simplify $\log_6 4 + \log_6 9$

4 EXPAND

Expand and simplify $(x - 2)(x + 2)(x^2 + 4)$

5 FACTORISE

Factorise $3x^2 + 10x - 8$

6 ALGEBRAIC FRACTIONS

Simplify $\dfrac{2}{x-1} - \dfrac{1}{x+1}$

7 LINEAR EQUATIONS & INEQUALITIES

Solve $\dfrac{x}{3} - \dfrac{x-2}{4} = 1$

8 QUADRATIC EQUATIONS

Solve $3x^2 - 5x - 2 = 0$

9 SIMULTANEOUS EQUATIONS

Find where $y = x^2$ and $y = x + 2$ meet.

10 FUNCTIONS ★

$f(x) = 2x + 1$ and $g(x) = x^2$. Find $f(g(3))$.

11 GRAPHS

State the equations of the asymptotes of $y = \dfrac{2}{x} + 1$.

12 POLYNOMIALS

Show that $(x + 2)$ is a factor of $x^3 + x^2 - 4x - 4$, then factorise fully.

13 LINES

Find the equation of the line through $(0,\ 4)$ perpendicular to $3x - y + 2 = 0$.

14 COORDINATE GEOMETRY

Show that $A(1,\ 1)$, $B(4,\ 5)$ and $C(8,\ 2)$ form a right-angled triangle.

15 TRIG RATIOS & EXACT VALUES ★

Convert $135°$ to radians, as an exact value.

16 SINE & COSINE RULES

Find the area of a triangle with sides 8 cm and 10 cm and an included angle of $30°$.

17 GEOMETRY

Chords AB and CD cross at P. $AP = 4$, $PB = 6$ and $CP = 3$. Find PD.

18 PROBABILITY

$P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Find $P(A \cup B)$ and $P(A \mid B)$.

19 FINANCIAL MATHS

\$10 000 is invested at 6% p.a., compounded monthly. Write an expression for its value after 3 years.

20 PROBLEM SOLVING

A ball's height is $h = 20t - 5t^2$ metres after t seconds. When does it land, and what is its greatest height?

BONUS Solve $2^{2x} - 6 \times 2^x + 8 = 0$

WEEK 4

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Indices, surds & logs · Algebra · Equations · Functions & graphs · Coordinate geometry · Trigonometry · Geometry & chance · In context · ★ Year 11 preview (Stage 6 Advanced) · No

1 INDICES

Simplify $(a^{\frac{1}{2}}b^{-1})^2 \times ab^3$

2 SURDS

Rationalise the denominator of $\dfrac{4}{\sqrt{7}-\sqrt{3}}$

3 LOGARITHMS

Solve $\log_2 x = 5$

4 EXPAND

Expand $(x+2)^3$

5 FACTORISE

Factorise $x^3 - 9x$

6 ALGEBRAIC FRACTIONS

Simplify $\dfrac{x^2-9}{2x+6} \times \dfrac{4}{x-3}$

7 LINEAR EQUATIONS & INEQUALITIES

Solve $-3 < 2x + 1 \leq 7$

8 QUADRATIC EQUATIONS

Find the discriminant of $2x^2 - 3x + 5 = 0$. How many real roots are there?

9 SIMULTANEOUS EQUATIONS

Solve simultaneously: $x + y = 7$ and $xy = 12$

10 FUNCTIONS

Is the relation $y^2 = x$ a function? Explain.

11 GRAPHS

Describe how to get the graph of $y = (x-3)^2 - 1$ from $y = x^2$.

12 POLYNOMIALS

$P(x) = x^3 - 6x^2 + 11x - 6$ and $P(1) = 0$. Factorise $P(x)$ fully and solve $P(x) = 0$.

13 LINES

Find where $2x + 5y = 20$ cuts the axes, and the area of the triangle it makes with the axes.

14 COORDINATE GEOMETRY

Write the equation of the circle with centre $(2,\ -3)$ and radius 5.

15 TRIG RATIOS & EXACT VALUES

Find the exact value of $\cos 150^\circ$

16 SINE & COSINE RULES

Find the largest angle of a triangle with sides 3, 5 and 7.

17 GEOMETRY

Find the sum of the interior angles of a decagon.

18 PROBABILITY

Two cards are drawn without replacement from ten cards numbered 1 to 10. Find $P(\text{both even})$.

19 FINANCIAL MATHS

Which earns more on \$1000 over 3 years: 12% p.a. simple interest or 10% p.a. compound interest?

20 PROBLEM SOLVING

The sum of the first n positive integers is $\frac{n(n+1)}{2}$. How many must be added to make 210?

BONUS Find k if $x^2 + kx + 9 = 0$ has exactly one real solution.

WEEK 5

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Indices, surds & logs · Algebra · Equations · Functions & graphs · Coordinate geometry · Trigonometry · Geometry & chance · In context · ★ Year 11 preview (Stage 6 Advanced) · No

1 INDICES

Write $\dfrac{\sqrt{x}}{x^2}$ as a single power of x.

2 SURDS

Find a and b if $(3+\sqrt{2})^2 = a + b\sqrt{2}$.

3 LOGARITHMS

Simplify $2\log_{10} 5 + \log_{10} 4$

4 EXPAND

Expand $\left(x + \dfrac{1}{x}\right)^2$

5 FACTORISE ★

Factorise $x^3 - 8$

6 ALGEBRAIC FRACTIONS

Simplify $\dfrac{1 - \frac{1}{x}}{1 - \frac{1}{x^2}}$

7 LINEAR EQUATIONS & INEQUALITIES

Solve $3(x-2) - 4(1-x) > 4$

8 QUADRATIC EQUATIONS

Solve $(x-3)^2 = 7$

9 SIMULTANEOUS EQUATIONS

Find where the circle $x^2 + y^2 = 25$ meets the line $y = x + 1$.

10 FUNCTIONS

$f(x) = 3x - 2$. Find $f(a+1) - f(a)$.

11 GRAPHS

Find the vertex of $y = x^2 - 6x + 5$ by completing the square.

12 POLYNOMIALS

Divide $x^3 - 4x^2 + x + 6$ by $(x-2)$. State the quotient and remainder.

13 LINES

Find the angle the line $y = \sqrt{3}x + 1$ makes with the positive x-axis.

14 COORDINATE GEOMETRY

Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

15 TRIG RATIOS & EXACT VALUES ★

θ is acute and $\sin\theta = \frac{3}{5}$. Use $\sin^2\theta + \cos^2\theta = 1$ to find $\cos\theta$.

16 SINE & COSINE RULES

How many triangles have $a = 6$, $b = 10$ and $A = 30°$? Explain.

17 GEOMETRY

PA and PB are tangents from P to a circle with centre O. If $\angle APB = 50°$, find $\angle AOB$.

18 PROBABILITY

In a class of 28, 15 study French, 12 study Japanese and 5 study both. A Japanese student is chosen at random. Find $P(\text{also studies French})$.

19 FINANCIAL MATHS

An investment grows to \$4840 after 2 years at 10% p.a., compounded annually. Find the amount invested.

20 PROBLEM SOLVING ★

For $y = x^2$, find the gradient of the line through $(2,\ 4)$ and $(2+h,\ (2+h)^2)$. What happens as $h \to 0$?

BONUS Solve $x - 5\sqrt{x} + 6 = 0$

WEEK 6

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Indices, surds & logs · Algebra · Equations · Functions & graphs · Coordinate geometry · Trigonometry · Geometry & chance · In context · ★ Year 11 preview (Stage 6 Advanced) · No

1 INDICES

Solve $4^{x+1} = 8^x$

2 SURDS

Simplify $\dfrac{(\sqrt{6}+\sqrt{2})(\sqrt{6}-\sqrt{2})}{\sqrt{2}}$

3 LOGARITHMS

Solve $2^x = 7$, giving an exact answer.

4 EXPAND

Expand $(2x-1)^3$

5 FACTORISE

Factorise $2x^3 + 3x^2 - 8x - 12$

6 ALGEBRAIC FRACTIONS

Solve $\dfrac{3}{x-2} = \dfrac{2}{x+1}$

7 LINEAR EQUATIONS & INEQUALITIES

For which values of x is $5 - 2x$ positive?

8 QUADRATIC EQUATIONS

For which values of k does $x^2 - 4x + k = 0$ have two distinct real roots?

9 SIMULTANEOUS EQUATIONS

The line $y = 2x + c$ touches the parabola $y = x^2$ at exactly one point. Find c.

10 FUNCTIONS ★

Is $f(x) = x^3 - x$ odd, even or neither? Show why.

11 GRAPHS

For $y = 2^x - 4$, find the y-intercept, the x-intercept and the asymptote.

12 POLYNOMIALS

$P(x) = x^3 + ax + 2$ leaves a remainder of 5 when divided by $(x-1)$. Find a.

13 LINES

Find the point of intersection of $3x - 2y = 4$ and $x + 4y = 6$.

14 COORDINATE GEOMETRY

Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at $(3,\ 4)$.

15 TRIG RATIOS & EXACT VALUES ★

Solve $2\cos\theta + 1 = 0$ for $0° \le \theta \le 360°$.

16 SINE & COSINE RULES

A ship sails 5 km on a bearing of $060°$T, then 8 km on a bearing of $180°$T. How far is it from its start?

17 GEOMETRY

Prove that the angle in a semicircle is a right angle.

18 PROBABILITY

A and B are independent, with $P(A) = 0.3$ and $P(B) = 0.4$. Find $P(A \cup B)$.

19 FINANCIAL MATHS

An account pays 4.8% p.a., compounded monthly. Find the monthly interest rate and the number of compounding periods in 5 years.

20 PROBLEM SOLVING ★

The tangent to $y = x^2$ at $x = a$ has gradient $2a$. Find the point where the tangent is parallel to $y = 6x - 1$.

BONUS Solve $\log_2 x + \log_2(x-2) = 3$

Hints & formulas

Indices, surds & logs

$a^{-n} = \frac{1}{a^n} \cdot a^{\frac{m}{n}} = (\sqrt[n]{a})^m$

$\frac{a}{\sqrt{b}} = \frac{a\sqrt{b}}{b}$; use the conjugate for $\sqrt{a} - \sqrt{b}$

$\log_a x = n \Leftrightarrow a^n = x$

$\log_a xy = \log_a x + \log_a y \cdot \log_a x^n = n \log_a x$

Algebra

$(a \pm b)^2 = a^2 \pm 2ab + b^2 \cdot (a + b)(a - b) = a^2 - b^2$

Factorise: HCF, DOTS, trinomial, grouping

★ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Algebraic fractions: factorise first, then cancel factors

Equations

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \cdot \Delta = b^2 - 4ac$

$\Delta > 0$: two roots · $\Delta = 0$: one · $\Delta < 0$: none

Inequalities: ×/÷ by a negative reverses the sign

Simultaneous: substitute or eliminate; a line and a curve can meet twice

Functions & graphs

$f(a)$: replace x with a · domain: the x-values allowed

Vertical line test: one y for each x

$y = (x - h)^2 + k$: vertex (h, k) · $y = \frac{k}{x}$: asymptotes $x = 0$, $y = 0$

Remainder theorem: remainder $= P(a)$ when dividing by $(x - a)$

★ Odd: $f(-x) = -f(x)$ · even: $f(-x) = f(x)$

Coordinate geometry

$y - y_1 = m(x - x_1)$ · perpendicular: $m_1 m_2 = -1$

$m = \tan\theta$ · general form $ax + by + c = 0$

Circle: $(x - a)^2 + (y - b)^2 = r^2$; complete the square to find the centre

Trigonometry

$\sin 30° = \frac{1}{2}$, $\cos 30° = \frac{\sqrt{3}}{2}$, $\tan 45° = 1$

$\sin(180° - \theta) = \sin\theta \cdot \cos(180° - \theta) = -\cos\theta$

$\frac{a}{\sin A} = \frac{b}{\sin B} \cdot c^2 = a^2 + b^2 - 2ab\cos C \cdot A = \frac{1}{2}ab\sin C$

★ $180° = \pi$ radians · $\sin^2\theta + \cos^2\theta = 1$

Geometry & chance

Cyclic quadrilateral: opposite angles add to $180°$

Tangent ⊥ radius · intersecting chords: $AP \cdot PB = CP \cdot PD$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

In context

$A = P(1 + r)^n$, with r and n per compounding period

Depreciation $S = V_0(1 - r)^n$

★ Gradient of a secant $= \frac{f(x + h) - f(x)}{h}$; let $h \to 0$ for the tangent

Answers

1 Week 1 answers

1 $4\frac{1}{2}$ $2^2 + \frac{1}{2}$

2 $3\sqrt{3}$ $5\sqrt{3} - 2\sqrt{3}$

3 4 $3^4 = 81$

4 $2x^2 + x - 15$ $2x^2 + 6x - 5x - 15$

5 $(x - 7)(x + 4)$ $-7 \times 4 = -28$, $-7 + 4 = -3$

6 $\frac{7x+5}{12}$ $\frac{4(x+2)+3(x-1)}{12}$

7 $x \leq -4$ $-3x \geq 12$; divide by -3 and reverse

8 $x = 3$ **or** $x = -5$ $(x + 5)(x - 3) = 0$

9 $x = 3, y = 5$ $x + 2x - 1 = 8$

10 10 $4 + 6$

11 $(2,\ 3)$; $x = 2$ vertex form $y = (x - h)^2 + k$

12 4 $8 - 8 + 4$

13 $m = 2, c = 3$ $2y = 4x + 6$

14 $(1,\ 5)$; **10 units** $\sqrt{36 + 64}$

15 $\frac{1}{2}$ $\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{3}}$

16 $c = 7$ $c^2 = 9 + 25 - 2(3)(5)\cos 120° = 34 + 15$

17 $105°$ opposite angles of a cyclic quadrilateral are supplementary

18 $\frac{1}{6}$ 3 + 2 + 1 = 6 outcomes out of 36

19 **$5408** $5000 \times 1.04^2 = 5000 \times 1.0816$

20 **12 and 14** $n(n + 2) = 168$

B **7** square both sides: $x^2 + 2 + \frac{1}{x^2} = 9$

2 Week 2 answers

1 $3x^2$ $\sqrt[3]{27} \times x^{6 \times \frac{1}{3}}$

2 $2\sqrt{5}$ $\frac{10\sqrt{5}}{5}$

3 -3 $10^{-3} = 0.001$

4 $x^3 + 1$ $x^3 - x^2 + x + x^2 - x + 1$

5 $2(x - 2)(x + 2)$ HCF 2, then difference of squares

6 $\frac{x}{x+4}$ $\frac{x(x-4)}{(x-4)(x+4)}$

7 $x < 8$ $2x - 1 < 15$

8 $x = 2 \pm \sqrt{5}$ $\frac{4 \pm \sqrt{16+4}}{2}$

9 $x = 2, y = 3$ add: $7x = 14$

10 $x \geq 3$ need $x - 3 \geq 0$

11 $x = 3, x = -2$; $y = -6$ $(x - 3)(x + 2) = 0$

12 -2 remainder theorem: $P(1) = 1 + 2 - 5$

13 $y = -2x + 5$ $y + 1 = -2(x - 3)$

14 **centre** $(1,\ -2)$, **radius 4** compare with $(x - a)^2 + (y - b)^2 = r^2$

15 $\theta = 60°$ **or** $120°$ $\sin(180° - 60°) = \sin 60°$

16 $b = 3\sqrt{2}$ $b = \frac{6 \sin 30°}{\sin 45°} = \frac{3}{\frac{\sqrt{2}}{2}}$

17 **135 cm³** volumes in the ratio $8 : 27$

18 $\frac{2}{5}$ $\frac{3}{5} \cdot \frac{2}{4} + \frac{2}{5} \cdot \frac{1}{4} = \frac{8}{20}$

19 **$21 675** $30\,000 \times 0.85^2 = 30\,000 \times 0.7225$

20 **8 cm by 5 cm** $x(13 - x) = 40$, $(x - 5)(x - 8) = 0$

B **multiply by** $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ $\frac{5+2\sqrt{15}+3}{5-3} = \frac{8+2\sqrt{15}}{2}$

Answers

3 Week 3 answers

1 $x = \frac{3}{2}$ $3^{2x} = 3^3$

2 $13 - 4\sqrt{3}$ $12 - 4\sqrt{3} + 1$

3 2 $\log_6 36$

4 $x^4 - 16$ $(x^2 - 4)(x^2 + 4)$

5 $(3x - 2)(x + 4)$ check: $3x^2 + 12x - 2x - 8$

6 $\frac{x+3}{x^2-1}$ $\frac{2(x+1)-(x-1)}{(x-1)(x+1)}$

7 $x = 6$ $4x - 3(x - 2) = 12$

8 $x = 2$ **or** $x = -\frac{1}{3}$ $(3x + 1)(x - 2) = 0$

9 $(2,\ 4)$ **and** $(-1,\ 1)$ $x^2 - x - 2 = 0$

10 19 $g(3) = 9$; $f(9) = 19$

11 $x = 0$ **and** $y = 1$ $y = \frac{2}{x}$ moved up 1

12 $(x + 1)(x - 2)(x + 2)$ $P(-2) = -8 + 4 + 8 - 4 = 0$

13 $y = -\frac{1}{3}x + 4$ that line has $m = 3$

14 $m_{AB} \times m_{BC} = -1$ $\frac{4}{3} \times (-\frac{3}{4}) = -1$: right angle at B

15 $\frac{3\pi}{4}$ $135 \times \frac{\pi}{180}$

16 **20 cm²** $\frac{1}{2} \times 8 \times 10 \times \sin 30°$

17 8 $AP \times PB = CP \times PD$

18 0.7; 0.5 $0.5 + 0.4 - 0.2$; $\frac{0.2}{0.4}$

19 $10\,000 \times 1.005^{36}$ 0.5% per month for 36 months

20 $t = 4$ **s; 20 m** $5t(4 - t) = 0$; max halfway, at $t = 2$

B $x = 1$ **or** $x = 2$ let $u = 2^x$: $u^2 - 6u + 8 = 0$, $u = 2$ or 4

4 Week 4 answers

1 a^2b $ab^{-2} \times ab^3$

2 $\sqrt{7} + \sqrt{3}$ $\frac{4(\sqrt{7}+\sqrt{3})}{7-3}$

3 $x = 32$ $x = 2^5$

4 $x^3 + 6x^2 + 12x + 8$ $(x^2 + 4x + 4)(x + 2)$

5 $x(x - 3)(x + 3)$ HCF x, then difference of squares

6 2 $\frac{(x-3)(x+3)}{2(x+3)} \times \frac{4}{x-3}$

7 $-2 < x \leq 3$ subtract 1, then divide by 2

8 $\Delta = -31$; **none** $9 - 40$

9 $(3,\ 4)$ **or** $(4,\ 3)$ $x(7 - x) = 12$, $x^2 - 7x + 12 = 0$

10 **No** $x = 4$ gives $y = 2$ and $y = -2$

11 **3 right, 1 down** vertex moves from $(0, 0)$ to $(3, -1)$

12 $(x - 1)(x - 2)(x - 3)$; $x = 1, 2, 3$ divide by $(x - 1)$: $x^2 - 5x + 6$

13 $(10,\ 0)$, $(0,\ 4)$; **20 units²** $\frac{1}{2} \times 10 \times 4$

14 $(x - 2)^2 + (y + 3)^2 = 25$ $(x - a)^2 + (y - b)^2 = r^2$

15 $-\frac{\sqrt{3}}{2}$ $\cos 150° = -\cos 30°$

16 $120°$ $\cos C = \frac{9+25-49}{30} = -\frac{1}{2}$

17 $1440°$ $(10 - 2) \times 180°$

18 $\frac{2}{9}$ $\frac{5}{10} \times \frac{4}{9}$

19 **12% simple ($360 vs $331)** $1000 \times 1.1^3 = 1331$

20 **20** $n(n + 1) = 420 = 20 \times 21$

B $k = 6$ **or** $k = -6$ $\Delta = k^2 - 36 = 0$

Answers

5 Week 5 answers

1 $x^{-\frac{3}{2}}$ $x^{\frac{1}{2}-2}$

2 $a = 11, b = 6$ $9 + 6\sqrt{2} + 2$

3 2 $\log_{10} 25 + \log_{10} 4 = \log_{10} 100$

4 $x^2 + 2 + \frac{1}{x^2}$ the middle term is $2 \times x \times \frac{1}{x}$

5 $(x-2)(x^2+2x+4)$ $a^3 - b^3 = (a-b)(a^2+ab+b^2)$

6 $\frac{x}{x+1}$ multiply top and bottom by x^2: $\frac{x(x-1)}{(x-1)(x+1)}$

7 $x > 2$ $7x - 10 > 4$

8 $x = 3 \pm \sqrt{7}$ $x - 3 = \pm\sqrt{7}$

9 $(3,\ 4)$ **and** $(-4,\ -3)$ $2x^2 + 2x - 24 = 0$

10 3 $3a + 1 - (3a - 2)$

11 $(3,\ -4)$ $y = (x-3)^2 - 4$

12 $x^2 - 2x - 3$, **remainder 0** check: $(x-2)(x^2-2x-3) = x^3 - 4x^2 + x + 6$

13 $60°$ $\tan\theta = \sqrt{3}$

14 **centre** $(3,\ -2)$**, radius 5** $(x-3)^2 + (y+2)^2 = 12 + 9 + 4$

15 $\frac{4}{5}$ $\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$

16 **two** $\sin B = \frac{10 \times 0.5}{6} = \frac{5}{6} < 1$, and B can be acute or obtuse

17 $130°$ tangents $\perp$ radii: $360 - 90 - 90 - 50$

18 $\frac{5}{12}$ conditional: 5 of the 12 Japanese students

19 **$4000** $P = 4840 \div 1.21$

20 $4 + h$**; it approaches 4** $\frac{(2+h)^2-4}{h} = \frac{4h+h^2}{h}$

B $x = 4$ **or** $x = 9$ let $u = \sqrt{x}$: $u^2 - 5u + 6 = 0$, $u = 2$ or 3

6 Week 6 answers

1 $x = 2$ $2^{2x+2} = 2^{3x}$

2 $2\sqrt{2}$ $\frac{6-2}{\sqrt{2}} = \frac{4}{\sqrt{2}}$

3 $x = \log_2 7$ definition of a logarithm

4 $8x^3 - 12x^2 + 6x - 1$ $(4x^2 - 4x + 1)(2x - 1)$

5 $(2x+3)(x-2)(x+2)$ $x^2(2x+3) - 4(2x+3)$

6 $x = -7$ $3(x+1) = 2(x-2)$

7 $x < \frac{5}{2}$ $5 - 2x > 0$

8 $k < 4$ $\Delta = 16 - 4k > 0$

9 $c = -1$ $x^2 - 2x - c = 0$; $\Delta = 4 + 4c = 0$

10 **odd** $f(-x) = -x^3 + x = -f(x)$

11 $y = -3$**;** $x = 2$**;** $y = -4$ $2^0 - 4 = -3$; $2^x = 4$

12 $a = 2$ $P(1) = 1 + a + 2 = 5$

13 $(2,\ 1)$ $x = 6 - 4y$: $18 - 14y = 4$

14 $3x + 4y = 25$ radius gradient $\frac{4}{3}$, so tangent gradient $-\frac{3}{4}$

15 $\theta = 120°$ **or** $240°$ $\cos\theta = -\frac{1}{2}$: 2nd and 3rd quadrants

16 **7 km** angle between the legs is $60°$: $c^2 = 25 + 64 - 40$

17 **it is half the angle at the centre** the angle at the centre is a straight angle, $180°$; half of it is $90°$

18 0.58 $0.3 + 0.4 - 0.3 \times 0.4$

19 **0.4%; 60** $4.8 \div 12$; 5×12

20 $(3,\ 9)$ $2a = 6$

B $x = 4$ $x(x-2) = 8$, so $x = 4$ or -2; reject -2